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Effect of Subtle Neurological Dysfunction on Response to Haloperidol Treatment in Schizophrenia

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Objective: The primary purpose of this study was to assess whether an interaction between subtle neurological impairment and haloperidol plasma level affects treatment response and, if so, the impact on negative symptoms in particular. **Method:** Forty-three schizophrenic and two schizoaffective inpatients diagnosed according to Research Diagnostic Criteria were given, at the end of a 1-week placebo period, a baseline evaluation consisting of the Brief Psychiatric Rating Scale (BPRS), Scales for the Assessment of Positive and Negative Symptoms, Quantified Neurological Scale, and the Simpson-Angus Scale for extrapyramidal side effects. Subjects were randomly assigned to one of three haloperidol plasma ranges and treated for 6 weeks. At the end point the BPRS, Scales for the Assessment of Positive and Negative Symptoms, and Simpson-Angus Scale were readministered. Multiple linear regressions were used to assess the extent to which the interaction between neurological abnormality and haloperidol plasma level predicted the end-point symptoms once the baseline symptoms, neurological abnormality, and haloperidol plasma level were accounted for. **Results:** Those patients with higher levels of overall abnormality on the Quantified Neurological Scale at baseline and with frontal dysfunction in particular, had, with increasing haloperidol plasma levels, more severe negative symptoms at end point. Neurological dysfunction was not related to end-point positive symptoms. The effect was specific to end-point negative symptoms and was independent of extrapyramidal side effects. **Conclusions:** If confirmed, these findings may indicate that relatively intact frontal function is needed for improvement in negative symptoms and that those patients with schizophrenia who have subtle neurological dysfunction should be treated with lower doses of neuroleptics.

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There is general agreement on the presence of neurological dysfunction in schizophrenia (1). However, there is a lack of consensus about its clinical relevance. Some investigators find neurological soft signs in both patients with schizophrenia and their first-degree relatives and hence hypothesize that soft signs are a "trait marker" for schizophrenia (2). Others find neurodysfunction to be related to either many clinical features (3) or to none and hence brand neurological

soft signs as not clinically relevant (4, 5). Patients with schizophrenia with a deficit-type syndrome and neurological impairment may be a separate group (6) with perhaps a different profile of response to neuroleptic treatment (6, 7).

Some links have been suggested between neurodysfunction and negative symptoms. A number of investigators have reported an association between structural brain abnormalities, neuroleptic nonresponsiveness (both for positive and negative symptoms), and negative symptoms (8-13). However, the underlying pathophysiology of negative symptoms remains unclear.

We hypothesize that neurological dysfunction affects response to neuroleptics. The purpose of this investigation was to assess how the symptoms of patients with varying degrees of subtle neurological dysfunction are altered by different haloperidol plasma levels. We tested the interaction of neurological dysfunction and haloperidol plasma level in determining end-point symptoms, particularly negative symptoms.

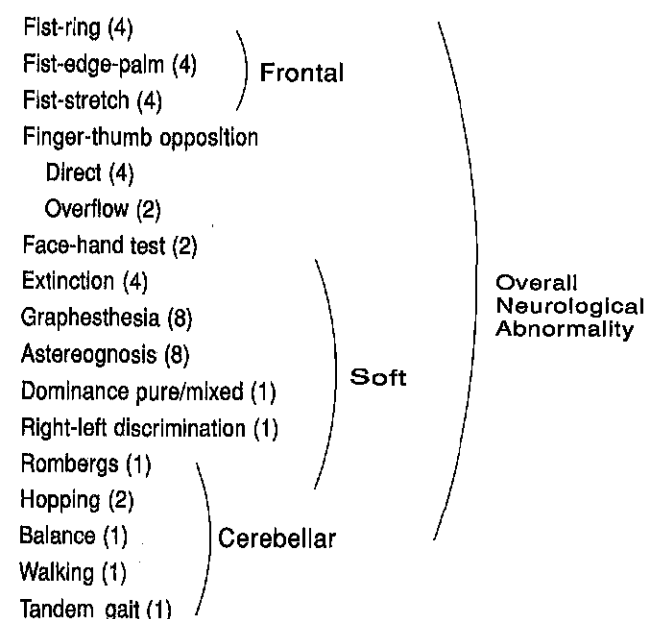
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FIGURE 1. Modified Quantified Neurological Scale Used in the Study of Neurological Dysfunction and Response to Haloperidol in Schizophrenia^a



^aNumbers in parentheses represent the number of items for each of the tests.

METHOD

Subjects

The cohort consisted of 45 patients, newly admitted to a large New York state hospital, usually by way of the emergency room of a large city hospital, who gave informed consent and met Research Diagnostic Criteria (RDC) (14) for schizophrenia or schizoaffective disorder. Diagnoses were established by using the Schedule for Affective Disorders and Schizophrenia (SADS) (15). The cohort consisted of the last 45 patients from a larger study designed to explore the relationship between plasma levels and clinical effects of haloperidol (16). The parent study reported only minor differences in clinical response among the three levels of haloperidol and thus suggested that low or moderate doses of neuroleptics are appropriate to treat many psychotic patients with schizophrenia. The 45 patients reported on here represent consecutive patients in the parent study (16) from the time the Quantified Neurological Scale (17) was added to the conclusion of that study. Table 1 presents demographic data on the cohort, including age, duration of illness, and psychiatric hospitalizations. There were 39 men (87%) and six women (13%) and 32 nonwhite subjects (71%) and 13 white subjects (29%). These demographic characteristics were essentially equivalent to those for the parent study. We studied 43 patients with schizophrenia and two patients with schizoaffective disorder. Of the patients with schizophrenia, 27 (63%) were paranoid type, 11 (26%) were undifferentiated, and five (11%) were other (disorganized or unspecified).

Procedure

The experiment had a 1-week, single-blind placebo period, which could be shortened for patients who could not tolerate the full week. Baseline ratings and haloperidol plasma levels were obtained on the last day of the placebo period. Our patient group had a placebo period of 6 days (SD=4) and an end-of-placebo haloperidol plasma level of 0.52 ng/ml (SD=1.6). After the placebo period, patients were entered into one of three randomly assigned, fixed, double-blind halo-

peridol plasma ranges: low (2–13 ng/ml), medium (13.1–24), or high (24.1–35); plasma ranges were monitored on a weekly basis and kept as close to the middle of the assigned range as possible (18). If a patient developed clinically significant extrapyramidal side effects during the study, he or she was placed on a regimen of benztropine. Of the 45 study patients, 40 (89%) received benztropine at some point during the treatment phase, and 34 (76%) were receiving benztropine at the time of the end-point rating. The only other medications allowed were chloral hydrate for insomnia and sodium amytal for extreme agitation. Nineteen patients (42%) received chloral hydrate at some point during the study period. None received sodium amytal. Patients received a comprehensive baseline evaluation at the end of the placebo period that included the Brief Psychiatric Rating Scale (BPRS) (19), the Scale for the Assessment of Negative Symptoms (20), the Scale for the Assessment of Positive Symptoms (21), a rating of extrapyramidal side effects (22), and the Quantified Neurological Scale (17).

Response Variables

The parent study (16) had two 6-week active treatment periods. The current report is based on data from the first treatment period (for which more data were available). The end point was generally at the end of treatment week 6, but it could be before then if the patient was discharged or refused to continue to participate in the study. At the end point scores on the BPRS, Scales for the Assessment of Positive and Negative Symptoms, and the Simpson-Angus Scale were again rated by trained researchers. Raters were trained in these different measures and showed good interrater reliability (16). Raters worked independently in pairs, and their item scores were averaged. In order to minimize the interrater variance for any one patient, as a rule the same raters evaluated each patient throughout the duration of the study. In our patient group there was at least one common rater between baseline and end-point evaluations in 41 (91%) of the patients. There were only four patients (9%) who had two different raters at the two time points. Raters of psychopathology were blind not only to haloperidol plasma level assignment but also to the results of the Quantified Neurological Scale. The total score on the Scale for the Assessment of Negative Symptoms as well as scores on the five subscales of the Scale for the Assessment of Negative Symptoms (affective flattening or blunting, alogia, avolition-apathy, anhedonia-asociality, and attention) (20) were evaluated. In psychopharmacological experiments of this type, one can use either the difference between the baseline and the end-point ratings or the end-point rating as the measure of response. We chose to use the end-point rating. For theoretical reasons outlined elsewhere (23), the results of using either of these two measures of response are identical for the nonconfounded variables; the only difference could be in the variance explained by the confounded variables (in our case, the baseline rating), which was not a variable of interest in this study.

Predictor Variables

Haloperidol plasma level was considered one of the predictors. The haloperidol plasma level was assayed by using a gas liquid chromatography system fitted with a nitrogen detector; the assay has been described elsewhere (16, 24). The haloperidol plasma level used in the analyses was the actual level obtained on the same day as the end-point rating.

Neurological abnormality was assessed by using the Quantified Neurological Scale. The level of abnormality on the Quantified Neurological Scale has been found to differentiate violent from nonviolent psychiatric inpatients (25–27) and to be related to their response to treatment (28). Since these original studies, we have expanded the Quantified Neurological Scale to 96 items, giving rise to 84 separate ratings. Thirty-six items, which assess cranial nerves, reflexes, cerebellar function, pyramidal tract function, and muscle strength, are used to screen for gross neurological dysfunction. The remaining items measure more subtle forms of neurological dysfunction and are the ones used for this report (figure 1 and appendix 1). (The administration and coding instructions are available on request from Dr. Convit.) The Quantified Neurological Scale is an instrument that

evaluates overall neurological abnormality, frontal, soft signs, and cerebellar function (figure 1 gives the makeup of each of these subscales). For the final analyses, all items on the scale were rated as 0 for normal and 1 for abnormal. The sum of item scores in each subscale reflected the abnormality score for that subscale. To assess reliability of the expanded Quantified Neurological Scale, two trained physicians independently rated 30 patients. For the individual items, kappa for the interrater reliabilities ranged from 0.69 to 1. All but three items (96%) had a kappa of 0.75 or greater. We attribute the high level of reliability to the very detailed administration and coding instructions that we have developed.

To assess the validity of the Quantified Neurological Scale, we contrasted, in a separate group of 28 psychiatric inpatients, the overall neurological abnormality scores with the Halstead-Reitan impairment index of the Halstead-Reitan neuropsychological battery (29) and the frontal score with the number of errors on the Wisconsin Card Sorting Test (30, 31), a widely accepted measure of frontal lobe function. The total Quantified Neurological Scale abnormality was significantly related to the Halstead-Reitan impairment index ($r=0.43$, $N=28$, $p=0.02$). The frontal abnormality subscore, which uses the newly added items, was significantly related to the number of errors on the Wisconsin Card Sorting Test ($r=0.47$, $N=28$, $p=0.01$).

Baseline symptoms have been reported to be related to the level of psychopathology at the end point of treatment. We therefore used it as one of the predictor variables.

Interaction between medication level and neurological abnormality score was the variable of primary interest in this study. We sought to determine the effect of this interaction once the effect of the other predictors on the end-point symptoms had been taken into account. We considered haloperidol plasma level, neurological abnormality score, and baseline symptoms as potential predictors for our dependent variable (end-point symptoms). This would allow us to assess how much additional variance would be accounted for by the interaction (overall neurological abnormality by haloperidol plasma level) after the variance explained by the other potential predictors had been taken into account.

Statistical Analyses

We used descriptive statistics and multiple linear regression. The regression was used to assess the extent to which the independent predictor variables (neurological abnormality score, baseline symptoms, haloperidol plasma level, and the interaction between plasma level and neurological abnormality) predict the dependent variable (level of symptoms at end point). All tests were two-tailed.

Separate regression equations were calculated by using the total neurological abnormality and the abnormality on the frontal, soft signs, and cerebellar subscales as independent variables. A total of 12 analyses were run: four (one for each of the four neurological subscales) for each of the end-point scores on the Scales for the Assessment of Positive and Negative Symptoms and BPRS. If these total end-point models for the three scales measuring symptoms were significant, then post hoc analyses were performed to examine the contribution of the independent variables in explaining the end-point subscales of the Scales for the Assessment of Positive and Negative Symptoms and the BPRS factors.

RESULTS

The patients studied were primarily men close to 30 years of age with a fairly chronic psychiatric illness (table 1). Most were nonwhite. Subjects were quite symptomatic, as indicated by the baseline BPRS total score, the total score on the Scale for the Assessment of Positive Symptoms, and the total score on the Scale for the Assessment of Negative Symptoms (table 1). The average overall neurological abnormality was 8.8, and the frontal abnormality was 4.4. The end-point haloperidol plasma level was 19.9 ng/ml (SD=9.9), which in our

TABLE 1. Characteristics of 45 Schizophrenic and Schizoaffective Inpatients

Variable	Mean	SD	Range
Age (years)	33.7	8.7	20–62
Hospitalizations ^a	7.6	4.1	1–15
Duration of illness (years) ^b	13.8	8.9	0–46
Baseline BPRS score	54.1	9.4	32–81
Baseline score on Scale for the Assessment of Positive Symptoms ^c	13.6	3.3	6–19
Baseline score on Scale for the Assessment of Negative Symptoms ^c	13.9	4.8	3–25
Total neurological abnormality score	8.8	5.6	0–27
Frontal abnormality score	4.4	3.3	0–12
End-point haloperidol level (ng/ml)	19.9	9.9	5–41

^aN=42.

^bN=41.

^cN=44.

patient group corresponded to a daily dose of 33.7 mg (SD=15.3) administered to a 70-kg man.

None of the models with end-point total score on the BPRS or the Scale for the Assessment of Positive Symptoms as the dependent variable showed a significant contribution for the Quantified Neurological Scale total abnormality score or any of the neurological subscales. The models with the end-point total score on the Scale for the Assessment of Negative Symptoms as the dependent variable did show a significant contribution from the total Quantified Neurological Scale and frontal abnormality scores. The post hoc analyses with the end-point affective flattening, alogia, and avolition-apathy subscales as dependent variables (table 2) were significant. The post hoc analyses with Scale for the Assessment of Negative Symptoms subscales of attention and anhedonia were not significant. It should be noted in table 2 that the residual R^2 (percent of variance explained independently by each of the predictor variables) for the interaction term (overall neurological abnormality by haloperidol plasma level) has a positive sign because the less the response, the higher the end-point symptoms. Because the affective flattening, alogia, and avolition-apathy post hoc models were significant, we also ran, as a cross-check, further analyses with the negative symptoms—factor 2—of the BPRS (32) as dependent variables. Although none of these BPRS factor 2 models was significant, the interaction terms were all positive, and the one with frontal abnormality as an independent variable showed a statistical trend for the interaction of frontal abnormality by haloperidol plasma level ($t=1.8$, $df=37$, $p=0.09$, partial $R^2=0.08$).

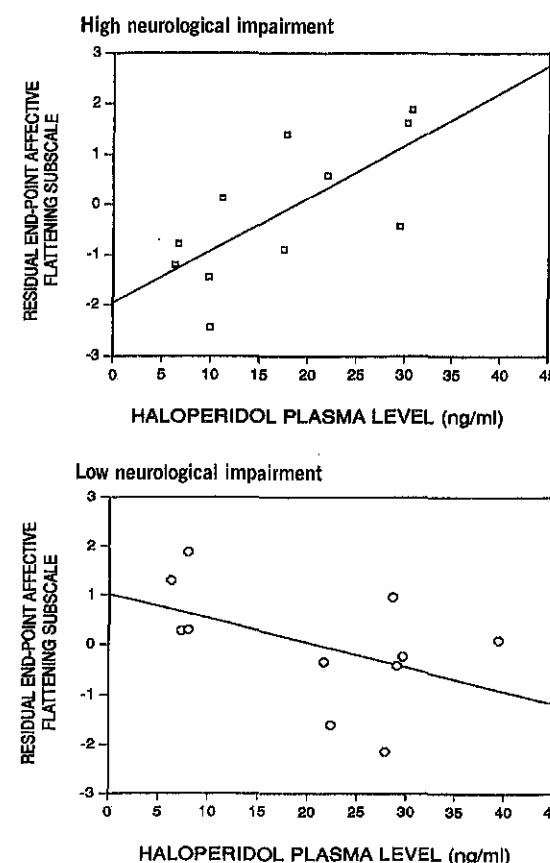
The frontal subscale had a significant interaction term (frontal abnormality by haloperidol plasma level) for the end-point total score model of the Scale for the Assessment of Negative Symptoms. In similar fashion to the total neurological abnormality, the post hoc analyses that used the frontal abnormality score as an independent variable resulted in significant and positive interaction terms (frontal abnormality by haloperidol plasma level) for the affective flattening, alogia, and avolition-apathy models (table 2). Again, the models for attention and anhedonia were not significant.

TABLE 2. Prediction of End Point in Total Score and Subscale Scores for the Scale for the Assessment of Negative Symptoms (SANS) for 45 Schizophrenic and Schizoaffective Inpatients: Multiple Regression Analyses With Total Neurological Abnormality and Frontal Abnormality

Independent Variable	Dependent Variable											
	Total			Affective Flattening Subscale			Alogia Subscale			Avolition-Apathy Subscale		
	T	p	R ² (%) ^a	T	p	R ² (%) ^a	T	p	R ² (%) ^a	T	p	R ² (%) ^a
Neurological abnormality ^b	1.70	0.10	-8	2.05	0.05	-10	1.89	0.07	-9	2.58	0.01	-15
Baseline SANS or subscale score	3.58	0.001	-27	3.43	0.001	-25	2.69	0.01	-17	5.41	0.0001	-44
Haloperidol level	1.88	0.07	-9	3.09	0.004	-21	2.40	0.02	-14	2.65	0.01	-16
Neurological abnormality by haloperidol level ^b	2.18	0.04	+12	2.90	0.006	+19	2.35	0.02	+13	2.70	0.01	+16
Frontal abnormality ^b	1.59	0.12	-7	1.98	0.06	-10	1.63	0.11	-7	1.80	0.08	-8
Baseline SANS or subscale score	3.53	0.001	-27	3.40	0.002	-24	2.48	0.002	-15	4.98	0.0001	-40
Haloperidol level	2.05	0.05	-11	3.23	0.003	-23	2.27	0.03	-12	2.17	0.04	-11
Frontal abnormality by haloperidol level ^b	2.62	0.01	+17	3.26	0.003	+23	2.34	0.02	+13	2.26	0.03	+12

^aA negative value signifies a negative relationship, and a positive value signifies a positive relationship.

^bN=39-42.

FIGURE 2. Neurological Impairment in Schizophrenia and Relationship Between Haloperidol Plasma Level and End-Point Scores on the Affective Flattening Subscale of the Scale for the Assessment of Negative Symptoms

None of the analyses for end-point Scale for the Assessment of Negative Symptoms (or any of its subscales) as the outcome was significant when the soft sign or cerebellar subscales of the Quantified Neurological Scale were used as independent variables.

To represent graphically the findings of the significant interaction (overall neurological abnormality by haloperidol plasma level), we split the patients into quartiles based on their level of overall neurological abnormality. For the two extreme groups we plotted the residual end-point Scale for the Assessment of Negative Symptoms affective flattening or blunting subscale (accounting first for the baseline score), which had a highly significant interaction term, against haloperidol plasma level. Figure 2 shows the plots for the groups with high and low overall neurological abnormality. For the group with high neurological impairment, there was a positive relationship between haloperidol plasma level and residual end-point affective flattening, indicating that for higher haloperidol plasma levels, this group of patients had higher end-point ratings of affective flattening, that is, a worse response. On the other hand, for the group with low overall neurological abnormality, there was a negative relationship between haloperidol plasma level and residual end-point affective flattening, indicating that for this group, increasing haloperidol plasma levels gave rise to lower end-point affective flattening, that is, a better response. The slopes for these two equations were different ($t=-3.34$, $df=18$, $p=0.004$). In an analogous fashion to when we split the subjects into quartiles of total neurological abnormality (figure 2), when we split the subjects into quartiles according to their frontal lobe dysfunction, we found the same kinds of relationships as those depicted in figure 2 for the two groups at the low and high ends of frontal dysfunction. The slopes for these two lines, fitted on the plots of residual end-point Scale for the Assessment of Negative Symptoms affective flattening or blunting subscale and haloperidol plasma levels, representing the subjects in the high and the low frontal dysfunction quartiles, were also different ($t=-2.67$, $df=18$, $p=0.02$).

Side effects, as measured by the Simpson-Angus Scale (22), may have an impact on the relationship between haloperidol plasma level and end-point negative symp-

toms. We therefore added the end-point Simpson-Angus rating as an independent variable to the significant models. This side effect term did not decrease the significance of the overall neurological abnormality by haloperidol plasma level interaction term in any of the models. Also related to side effects, 34 patients were receiving benzotropine at the time of the end-point rating. The end-point score on the Scale for the Assessment of Negative Symptoms was not significantly different for those patients who were or were not receiving benzotropine ($F=0.02$, $df=1$, 37 , $p=0.88$). Four of the 34 patients receiving benzotropine and two of the 11 other patients had missing end-point scores on the Scale for the Assessment of Negative Symptoms. For the 30 subjects with end-point scores on the Scale for the Assessment of Negative Symptoms who were receiving benzotropine at the end-point rating, the benzotropine dose was not related to the scale score (Pearson $r=0.10$, $N=30$, $p=0.57$).

Nineteen patients received chloral hydrate for insomnia at some point in the active treatment phase; 26 did not. There were missing end-point scores on the Scale for the Assessment of Negative Symptoms for one of the 19 patients receiving chloral hydrate and five of the 26 other patients. There were no significant differences in end-point scale scores for the patients who did or did not receive chloral hydrate ($F=0.26$, $df=1$, 37 , $p=0.61$).

CONCLUSIONS

Some authors have described an association between negative symptoms and brain dysfunction or structural abnormality (6, 10-13, 33-35). Others have described a relationship between higher haloperidol levels and worsening of negative symptoms (36). However, the underlying pathophysiology for this effect has not been elucidated. Many clinicians believe that when a patient is grossly compromised neurologically, a conventional neuroleptic may give rise to intolerable levels of side effects and actually make the patient symptomatically worse. Little evidence exists to date on how more subtle forms of neurodysfunction can affect the response to neuroleptics in schizophrenia and whether this relationship, if it exists, is dose related. This report, to our knowledge, presents the first evidence that worsening of negative symptoms with increasing plasma levels of haloperidol may be associated with subtle neurological dysfunction.

The most notable finding of our investigation was that higher plasma levels of haloperidol may worsen the negative symptoms of those patients who are neurologically impaired. Namely, there is a significant interaction between haloperidol plasma level and the neurological dysfunction in determining their negative symptoms at end point. These findings were equally strong for the overall neurological impairment and the specific frontal impairment. The cerebellar and soft signs subscales showed no statistical significance or

trend. This prompted us to conclude that the frontal abnormalities, which are also part of the total abnormality score, were probably the main contributor to the effect seen for the overall abnormality.

The fact that we did not find a significant relationship between the overall neurological abnormality by haloperidol plasma level or the frontal abnormality by haloperidol plasma level interaction terms and the end-point BPRS negative symptoms (factor 2) was not surprising to us. Although the BPRS factor 2 score is related to the score on the Scale for the Assessment of Negative Symptoms (variance shared by them is 64% in our data set), it is made up of only four items and is thus not expected to be as sensitive as the Scale for the Assessment of Negative Symptoms, which has broader variability. Moreover, the overlap in the variance between BPRS factors and the subscales of the Scale for the Assessment of Negative Symptoms is only about 50% (37). It should be noted, however, that for factor 2, these interaction terms were also positive in direction in our data set and showed a trend for the frontal model.

In general, persons with neuropsychiatric impairments caused by coarse brain disease are more vulnerable to the toxic effects of centrally active drugs (38, 39). Our data show that this greater vulnerability may apply even to schizophrenic patients with subtle degrees of neurological dysfunction. The relationship between subtle neurological dysfunction and haloperidol plasma level in determining the end-point negative symptoms is moderately strong: the interaction terms of overall neurological abnormality by haloperidol plasma level and frontal abnormality by haloperidol plasma level explained 12%-19% and 12%-23%, respectively, of the residual variance for the affective flattening, alogia, and avolition-apathy subscales of the Scale for the Assessment of Negative Symptoms. This is to say that an additional 12%-19% and 12%-23% were explained by each of the interaction terms after the independent contributions of the baseline negative symptom rating, haloperidol plasma level, and overall and frontal abnormalities were taken into account. Our study is strengthened by the fact that the diagnoses were established by using the SADS (15) and the RDC (14), the neurological dysfunction was established in a relatively medication-free state by using a reliable and valid neurological assessment, and the amount of haloperidol in plasma was established. Although the only measure of neurocognitive function was the Quantified Neurological Scale, we have shown that the scores on this scale are significantly related to other commonly used and validated methods of neurocognitive assessment.

Negative symptoms generally improve when treated with clinical doses of D₂ blockers (40, 41). However, high doses of D₂ blockers can elicit side effects that are phenomenologically difficult to distinguish from negative symptoms (42, 43). Most of our patients were taking benzotropine to prevent those side effects, and we also demonstrated that side effects, as determined by

the Simpson-Angus Scale, did not affect our findings. Considerable phenomenological similarities exist between frontal lobe syndromes and negative symptoms (44). Frontal lobes may be related to the genesis of negative symptoms or are needed for their improvement (45). When we split the subjects into quartiles according to their frontal lobe dysfunction, we found that the patients in the high frontal dysfunction group had higher end-point negative symptoms with increasing haloperidol plasma level, whereas for the low frontal impairment group, the opposite was true. These are the same kinds of relationships as those depicted in figure 2. This suggests to us that a frontal lobe mechanism may be responsible for the interaction effect between neurodysfunction and haloperidol plasma level on the end-point negative symptoms.

We studied only a relatively small number of patients (N=45), and we do not know whether these findings are generalizable to less chronic schizophrenic populations. If confirmed, these findings have considerable potential practical clinical implications.

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APPENDIX 1. Neurologic Scale

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Name of Patient _____	Date of Hospital Admission _____
Patient ID _____	Date of Test _____ (Month/Day/Year)
Current Neuroleptic _____	Daily Dose _____
Other meds _____	Cogentin Current Daily Dose _____

NEUROLOGICAL SCREEN:		
Pupillary Reaction: _____	Nystagmus: _____	Extraocular Movements: _____
0=Normal	0=Absent	0=Normal
1=Abnormal	1=Suggestive	1=Abnormal
	2=Present	

Visual Field: Single stimulation _____	Double Stimulation _____
0=Normal	
1=Abnormal	

Facial Asymmetry: _____	Hearing: <i>Rinne</i> _____	<i>Weber</i> _____
0=Absent	0=Normal	
1=Present	1=Right	
	2=Left	

Tongue Asymmetry: _____	
1=Yes	
1=Abnormal	

Walking Associated Symmetry: _____	Muscle Strength Asymmetry: _____
0=Normal	0=Normal
1=Abnormal	1=Abnormal

Arm Drift Present: _____	
0=No	
1=Yes	

Tendon Reflex Asymmetry: _____	Tendon Reflexes Amplitude: _____
0=No	0=(+1) or (+2)
1=Yes	1=Areflexia
	2=Hyperreflexia

Clonus: _____	Right _____	Left _____
0=No		
1=Yes		

Plantar Response (Babinski): _____	Right _____	Left _____
0=No		
1=Yes		

Heel to Shin: _____	Right _____	Left _____
0=Normal		
1=Abnormal		

Finger to Nose: _____	Eyes Open Right _____	Eyes Open Left _____
0=Normal	Eyes Closed Right _____	Eyes Closed Left _____
1=Suggestive		
2=Abnormal		

Pronation-Supination: _____	One Hand Right _____	Overflow Right _____
0=Normal	One Hand Left _____	Overflow Left _____
1=Suggestive	Two Hands Right _____	Two Hands Left _____
2=Abnormal		

Foot Taps: _____	One Foot Right _____	Overflow Right _____
0=Normal	One Foot Left _____	Overflow Left _____
1=Suggestive	Two Feet Right _____	Two Feet Left _____
2=Abnormal		

SOFT SIGNS AND FRONTAL ITEMS:			
Dominance: L=1, R=2	Comb _____	Write _____	Throw _____
	Telescope _____	Step on roach _____	
		Brush _____	Hole _____

R-L Discrimination:	Right Hand _____	Left Hand _____	Right Hand to Right Ear _____
0=Correct	Left Hand to Left Ear _____	Right Hand to Left Ear _____	
1=Incorrect	Left Hand to Right Ear _____	Identification Right and Left on Examiner _____	
Hopping:	Right _____	Left _____	
0=Normal			
1=Abnormal			
Standing Balance: _____	Walking: _____	Rombergs: _____	
0=Normal	0=Normal	0=Normal	
1=Abnormal	1=Abnormal	1=Abnormal	
Tandem Gait: _____			
0=Normal			
1=Abnormal			
Finger Thumb Opposition:	One Hand Right _____	One Hand Left _____	
0=Normal	Overflow Right _____	Overflow Left _____	
1=Suggestive	Two Hands Right _____	Two Hands Left _____	
2=Yes			
Extinction:	RF - LH _____	LF - RH _____	F - F _____ H - H _____
0=Correct			
1=Extinguishes Left			
2=Extinguishes Right			
3=Irrelevant Answer			
Graphesthesia:	3 Right _____	8 Left _____	1 Right _____ 7 Left _____
0=Correct	8 Right _____	1 Left _____	7 Right _____ 3 Left _____
1=Incorrect			
Astereognosis:	Key Right _____	Quarter Left _____	Penny Right _____ Comb Left _____
0=Correct	Key Left _____	Quarter Right _____	Penny Left _____ Comb Right _____
1=Incorrect			
Face-Hand Test:	Right Face-Right Hand _____	Left Face-Left Hand _____	
Fist-Ring:	Left Arm Imitate _____	Left Arm Continue _____	
0=Normal	Right Arm Imitate _____	Right Arm Continue _____	
1=Suggestive			
2=Abnormal			
Fist-Edge-Palm:	Left Arm Imitate _____	Left Arm Continue _____	
0=Normal	Right Arm Imitate _____	Right Arm Continue _____	
1=Suggestive			
2=Abnormal			
Fist-Stretch:	Left Arm Imitate _____	Left Arm Continue _____	
0=Normal	Right Arm Imitate _____	Right Arm Continue _____	
1=Suggestive			
2=Abnormal			

Preliminary Application of Magnetic Resonance Spectroscopy to Investigate Lactate-Induced Panic

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Objective: To characterize changes associated with lactate-induced panic, proton magnetic resonance spectroscopy (MRS) was used to measure brain lactate during intravenous infusion of 0.5-M sodium lactate in panic disorder patients and comparison subjects. **Method:** Eight panic disorder subjects, five medicated and three unmedicated, and eight healthy comparison subjects were studied at baseline, during lactate infusion (5 meq/kg over 20 minutes), and after infusion. Localized proton MRS was used to acquire averaged spectra every 5 minutes from a 27-ml sampling volume in the insular cortex and adjacent regions. Brain lactate levels, quantitatively estimated in relationship to N-acetyl aspartate, were compared to blood lactate levels. **Results:** The procedure was generally well tolerated; one panic subject requested early termination before lactate infusion. Significant rises in brain lactate levels occurred for all subjects during infusion. The panic patients who responded to lactate (N=3) had significantly higher brain lactate levels before, during, and after infusion than did the comparison subjects (N=8) and medicated patients who were lactate nonresponders (N=4). After infusion the panic patients with lactate-induced panic exhibited a striking dissociation between decreasing blood lactate and further increases in brain lactate levels. **Conclusions:** These preliminary observations indicate that brain lactate increases during a standard lactate infusion. Lactate-induced panic is associated with greater increases than in comparison subjects and with prolonged elevations in brain lactate that are decoupled from falling blood lactate levels after completion of lactate infusion. Further investigation is necessary to clarify the mechanism(s) responsible for these findings and establish whether a causal relationship to the occurrence of lactate-induced panic exists.

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Panic attacks can be acutely precipitated in susceptible individuals by certain chemical stimuli. Sodium lactate, one precipitating agent, has been extensively investigated in an attempt to understand the pathophysiology of panic. Clinical studies of intravenous infusion of 0.5-M or 1-M sodium lactate suggest that lactate responsiveness is a sensitive and specific marker for panic (1). Although it is one of the most

replicated findings in biological psychiatry, the mechanism responsible for lactate-induced panic is unknown.

Panic patients appear to have underlying metabolic abnormalities that are accentuated during lactate infusion. Pitts and McClure's rationale for first infusing lactate (2) was based, in part, on observations of elevated lactate production during moderate exercise in individuals experiencing what then was termed "anxiety neurosis" (3-5). Panic patients also appear to eliminate lactate more slowly than normal comparison subjects, resulting in higher blood lactate levels (6) and more rapid increases (7) in response to the same lactate load during lactate infusion. This excessive rise in blood lactate among panic patients who panic during lactate infusion has been observed to normalize with repeated infusion after successful treatment with imipramine (6). However, peripheral measurements suggesting abnormal metabolic handling of lactate loads in panic patients during lactate infusion do not necessarily reflect what is occurring behind the blood-brain barrier (8).

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